

Smart Ticket & Issue Management System

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Problem Statement

Inefficient Tracking: Reliance on manual email- and spreadsheet-based ticket tracking methods.

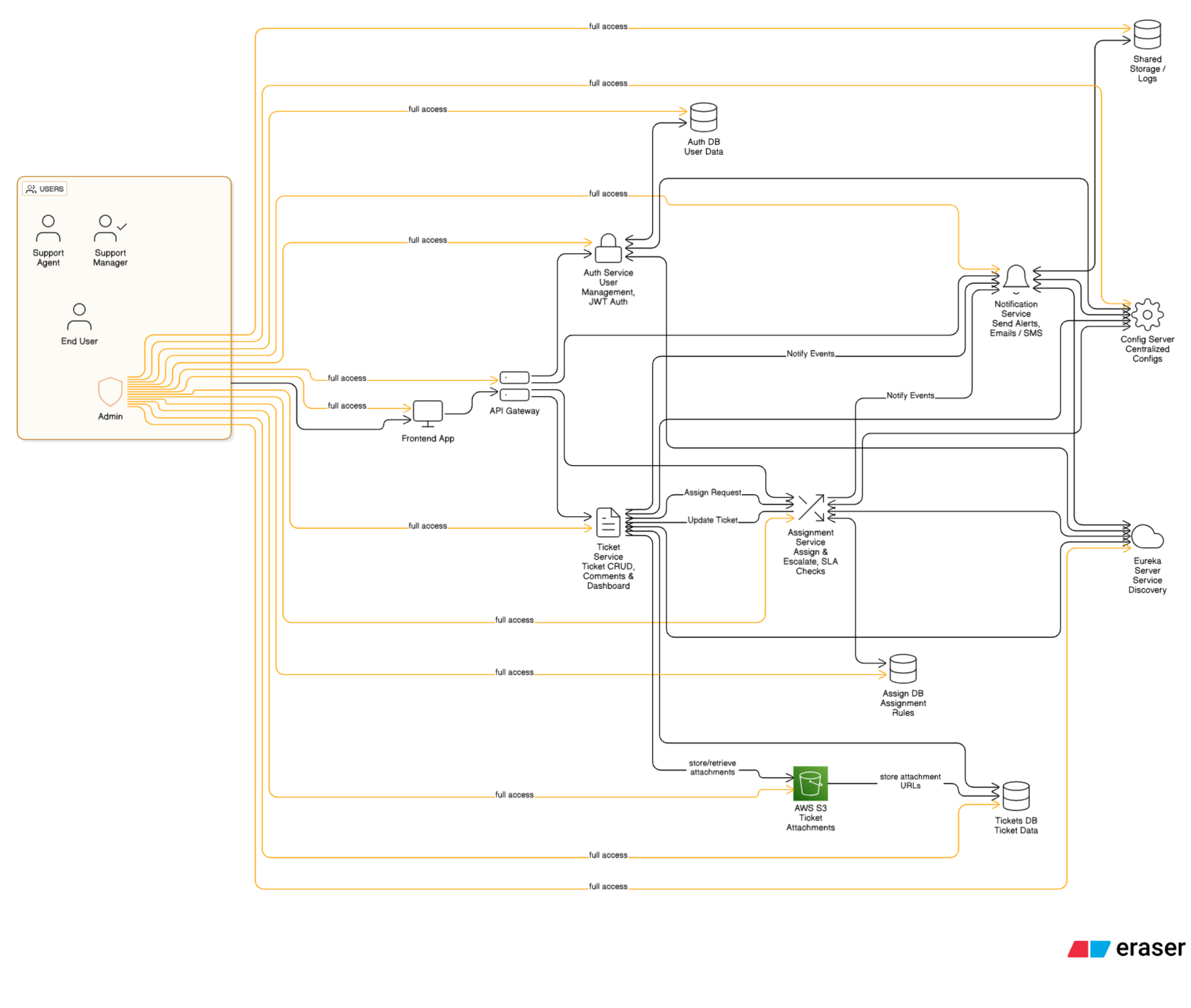
Decentralized Processes: Lack of a centralized system for automated ticket assignment and escalation.

Operational Blind Spots: Limited visibility into real-time ticket status, priority levels, and total resolution time.

Communication Gaps: Poor coordination between frontline support agents and management teams.

Performance Metrics: Minimal reporting capabilities regarding SLA (Service Level Agreement) compliance and overall team performance.

System Overview



Microservices & TechStack

6 Independent Microservices

- Auth Service (Authentication & Authorization)
- Ticket Service (Core ticket management)
- Assignment Service (Smart ticket routing)
- Notification Service (Real-time alerts)
- API Gateway (Request routing)
- Eureka Server (Service discovery)

Key Technologies

- Spring Boot & Spring Cloud
- PostgreSQL & MongoDB
- RabbitMQ (Message Broker)
- Docker & Docker Compose
- JWT Authentication

DataBase Design

- PostgreSQL (Relational Data)
 - Users Table: Authentication & user profiles
 - Assignments Table: Ticket assignment history
 - Notifications Table: Event notifications
 - Agent Metrics Table: Performance tracking
- MongoDB (Document Data)
 - Tickets Collection: Flexible schema for complex tickets
 - Comments: Array of comments within ticket
 - Attachments: File URLs and metadata
 - Custom Fields: Extensible ticket properties

Deployment

